

# **Mediterranean diet or extended fasting's influence on changing the intestinal microflora, immunoglobulin A secretion and clinical outcome in patients with rheumatoid arthritis and fibromyalgia: an observational study**



Background Alterations in the intestinal bacterial flora are believed to be contributing factors to many chronic inflammatory and degenerative diseases including rheumatic diseases. While microbiological fecal culture analysis is now increasingly used, little is known about the relationship of changes in intestinal flora, dietary patterns and clinical outcome in specific diseases. To clarify the role of microbiological culture analysis we aimed to evaluate whether in patients with rheumatoid arthritis (RA) or fibromyalgia (FM) a Mediterranean diet or an 8-day fasting period are associated with changes in fecal flora and whether changes in fecal flora are associated with clinical outcome. Methods During a two-months-period 51 consecutive patients from an Integrative Medicine hospital department with an established diagnosis of RA (n = 16) or FM (n = 35) were included in the study. According to predefined clinical criteria and the subjects' choice the patients received a mostly vegetarian Mediterranean diet (n = 21; mean age 50.9 +/-13.3 y) or participated in an intermittent modified 8-day fasting therapy (n = 30; mean age 53.7 +/- 9.4 y). Quantitative aerob and anaerob bacterial flora, stool pH and concentrations of secretory immunoglobulin A (sIgA) were analysed from stool samples at the beginning, at the end of the 2-week hospital stay and at a 3-months follow-up. Clinical outcome was assessed with the DAS 28 for RA patients and with a disease severity rating scale in FM patients. Results We found no significant changes in the fecal bacterial counts following the two dietary interventions within and between groups, nor were significant differences found in the analysis of sIgA and stool pH. Clinical improvement at the end of the hospital stay tended to be greater in fasting vs. non-fasting patients with RA (p = 0.09). Clinical outcome was not related to alterations in the intestinal flora. Conclusion Neither Mediterranean diet nor fasting treatments affect the microbiologically assessed intestinal flora and sIgA levels in patients with RA and FM. The impact of dietary interventions on the human intestinal flora and the role of the fecal flora in rheumatic diseases have to be clarified with newer molecular analysis techniques. The potential benefit of fasting treatment in RA and FM should be further tested in randomised trials.